What is Hyper-V?

Essential Guide for System Administrators

> **Hyper-V is Microsoft's virtualization platform** that allows you to run multiple operating systems simultaneously on a single physical computer by creating isolated virtual machines (VMs).

Core Concept

Hyper-V is a **Type 1 hypervisor** (runs directly on hardware) that creates virtual computers inside your physical computer. Each virtual machine acts like a separate, independent computer with its own operating system, applications, and resources.

Key Terms

- **Hypervisor:** Software that creates and manages VMs
- **Host:** Physical computer running Hyper-V
- **Guest:** Operating system inside a VM
- **Virtual Machine:** Software-based computer

Where You'll Find It

- **Windows Server:** Built-in role
- **Windows 10/11 Pro:** Optional feature
- **Hyper-V Server:** Free standalone version
- **Azure:** Powers Microsoft's cloud

Why System Admins Use Hyper-V

Server Consolidation

Run multiple servers on one physical machine, saving hardware costs and space.

Testing & Development

Create isolated environments for testing without affecting production systems.

Disaster Recovery

Easily backup, restore, and migrate entire server environments.

Legacy Application Support

Run older operating systems and applications on modern hardware.

Essential Features

- **Snapshots:** Save VM state at specific points in time
- **Live Migration:** Move running VMs between servers
- **Dynamic Memory:** Automatically adjust RAM allocation
- **Virtual Networking:** Create isolated network segments
- **Integration Services:** Enhanced VM performance
- **PowerShell Management:** Automate VM operations

Quick Start Requirements

- 64-bit processor with virtualization support (Intel VT-x or AMD-V)

- Minimum 4 GB RAM (8+ GB recommended)
- Windows Server or Windows 10/11 Pro/Enterprise
- Administrator privileges to install and configure

> **Bottom Line:** Hyper-V lets you maximize hardware usage, create flexible testing environments, and build resilient IT infrastructure by running multiple virtual computers on a single physical machine.
